

$$\nabla \cdot \vec{D} = \rho_v \quad \leftarrow \text{free charge}$$

Lecture 10

- Dielectrics

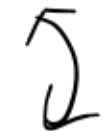
Materials

Depending on how materials respond to an external electric field, we classify them into two categories: **dielectrics and conductors**.

The key distinction is whether the material contains **free charges** that can readily move in response to an applied electric field.

* Perfect [electric] conductor (PEC) $\sigma \rightarrow \infty$

* Perfect dielectric = $\sigma = 0$



Materials

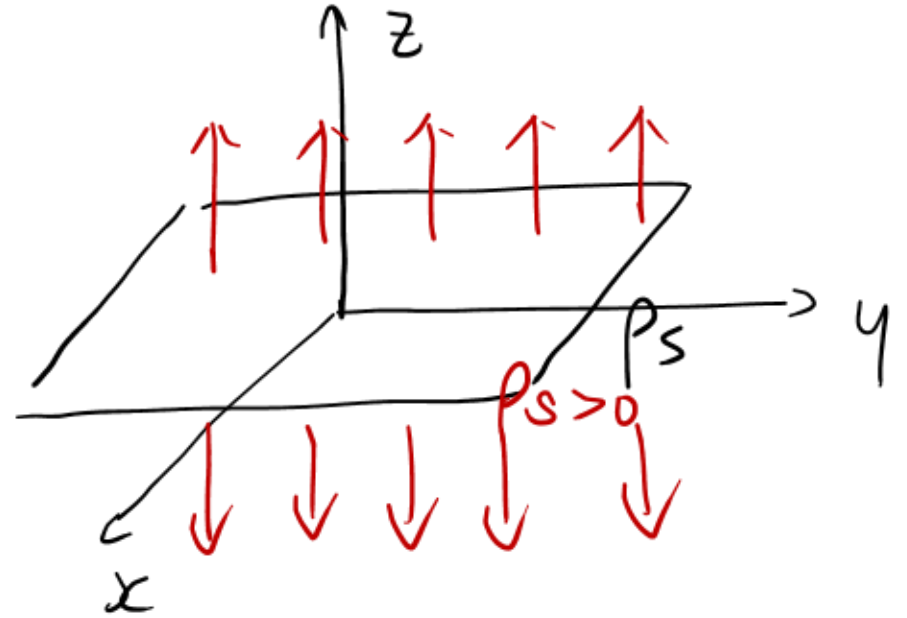
Material	Conductivity σ (S/m)
Conductors	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Tungsten	1.8×10^7
Zinc	1.7×10^7
Brass	1.5×10^7
Iron	10^7
Bronze	10^7

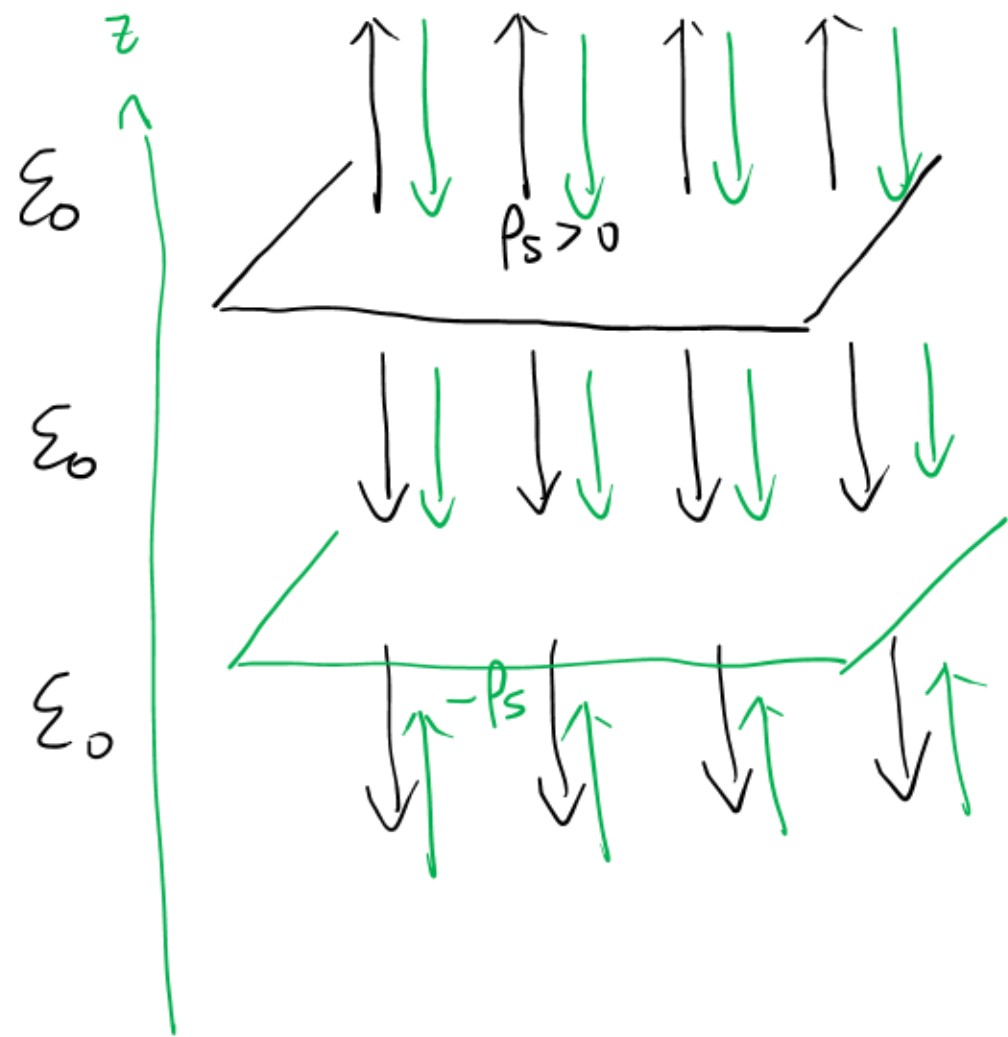
Material	Conductivity σ (S/m)
Air	10^{-15} to 10^{-14}
Glass	10^{-12}
Hard rubber	10^{-15}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}
Wax	10^{-17}

Dielectrics

Electric field from an infinite sheet with uniform surface charge density ρ_s ($z=0$)

$$\vec{D} = \begin{cases} \hat{z} \frac{\rho_s}{2} & z > 0 \\ -\hat{z} \frac{\rho_s}{2} & z < 0 \end{cases}$$





$$\textcircled{1} \quad \vec{D} = 0 \quad \vec{E} = 0$$

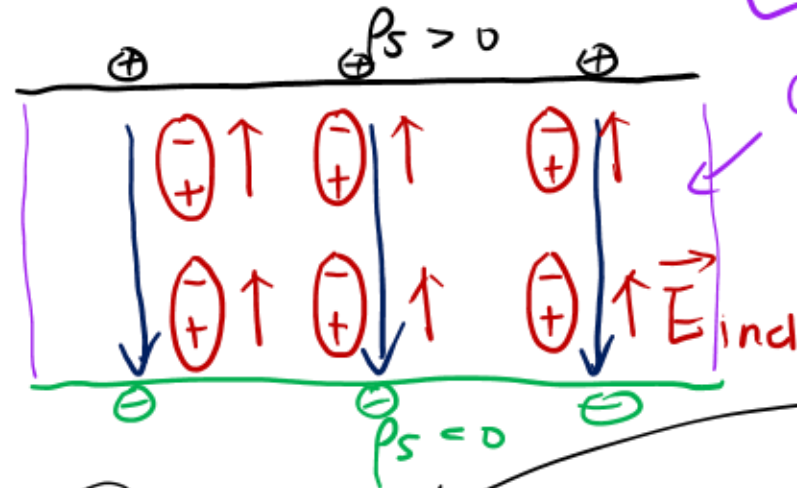
$$\textcircled{2} \quad \vec{D} = -\hat{z} \rho_s \quad \vec{E} = \frac{\vec{D}}{\epsilon_0} = -\hat{z} \frac{\rho_s}{\epsilon_0}$$

$$\textcircled{3} \quad \vec{D} = 0 \quad \vec{E} = 0$$

$$\vec{D} = \epsilon \vec{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e) \vec{E}_{total}$$

$$\vec{D}_{ext} = -\hat{z} \rho_s \quad \vec{E}_{ext} = -\hat{z} \frac{\rho_s}{\epsilon_0}$$



dielectric material

free charge

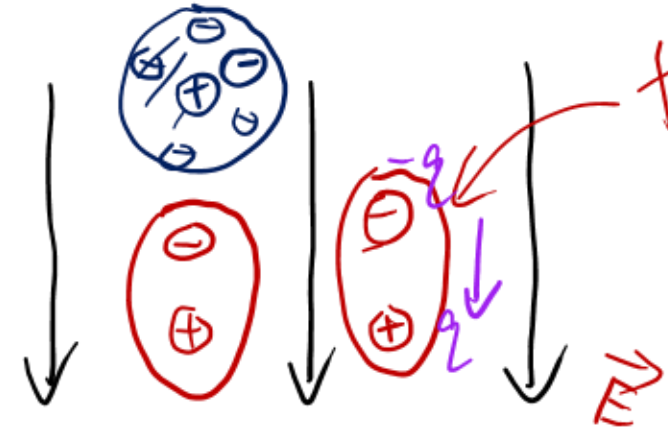
polarization of bound charge

$$\vec{E}_{tot} = \vec{E}_{ext} + \vec{E}_{ind}$$

free charge

point in oppo dir.

dielectric



form a small dipole moment

$\vec{p}$

$$|\vec{p}| = q \cdot d \quad [C \cdot m]$$

$$\vec{E}_{tot} = \vec{E}_{ext} + \vec{E}_{ind}$$

$$\epsilon_0 \vec{E}_{ext} = \epsilon_0 \vec{E}_{tot} - \epsilon_0 \vec{E}_{ind}$$

$$\vec{D}_{ext} = \epsilon_0 \vec{E}_{tot} + \vec{P}$$

$[C/m^2] \quad [E/m] \quad [V/m] \quad [C/m^2]$

⇒ Polarization of dielectric leads to net E-field reduction.

$$\vec{P} = -\epsilon_0 \vec{E}_{ind}$$

$$\vec{D}_{ext} = \epsilon_0 \vec{E}_{tot} + \chi_e \epsilon_0 \vec{E}_{tot} = \epsilon_0 (1 + \chi_e) \vec{E}_{tot}$$

$$\vec{P} = \chi_e \cdot \epsilon_0 \vec{E}_{tot}$$

electronic susceptibility, no unit

$$\vec{E}_{\text{total}} = \vec{E}_{\text{ext}} + \vec{E}_{\text{ind}}$$

$$\vec{E}_{\text{ext}} = \vec{E}_{\text{total}} - \vec{E}_{\text{ind}}$$

$$\epsilon_0 \vec{E}_{\text{ext}} = \epsilon_0 \vec{E}_{\text{total}} - \epsilon_0 \vec{E}_{\text{ind}}$$

$$\vec{D} = \epsilon_0 \vec{E}_{\text{total}} + \vec{P}$$

affected by both free charge and bound charge

$$= \epsilon_0 (1 + \chi_e) \vec{E}_{\text{total}}$$

ϵ_r relative permittivity

$$= \underbrace{\epsilon_0 \epsilon_r}_{\epsilon} \vec{E}_{\text{total}}$$

$$\boxed{\vec{D} = \epsilon \vec{E}}$$

Polarization Field

generated by bound charge

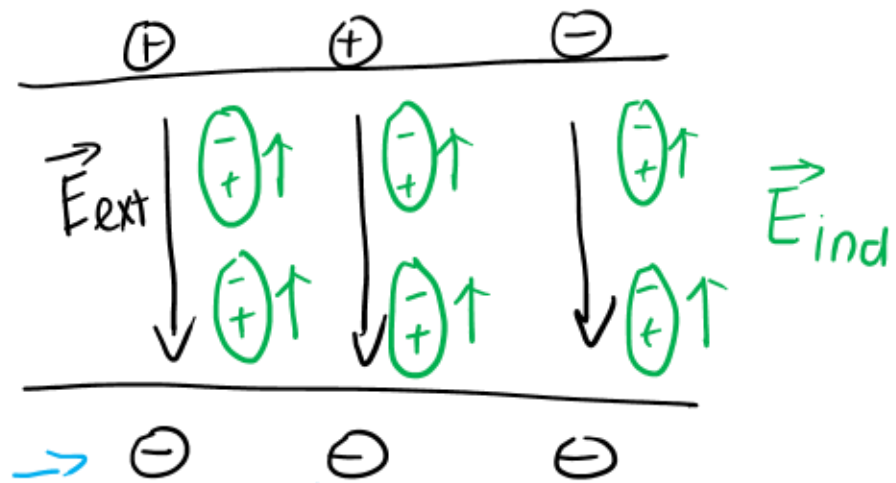
$$\vec{P} = -\epsilon_0 \vec{E}_{\text{ind}} \left[\frac{C}{m^2} \right]$$

its dir is along $\vec{E}_{\text{ext}}$.
its magnitude is the dipole moment per unit volume

$\vec{P}$ is usually compared to $\epsilon_0 \vec{E}_{\text{total}}$ and written as

$$\vec{P} = \chi_e \epsilon_0 \vec{E}_{\text{total}}$$

electric susceptibility [no unit]



Dielectrics

TABLE B-1

RELATIVE PERMITTIVITY ϵ_r
OF COMMON MATERIALS^a

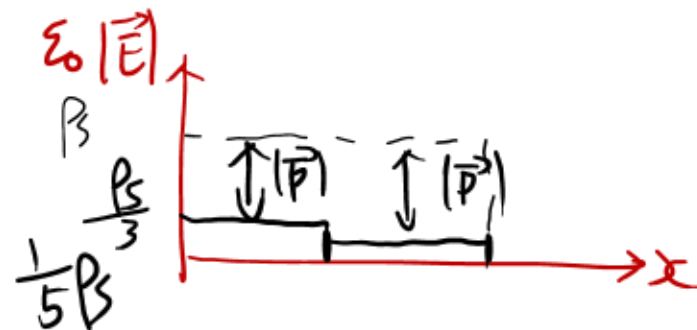
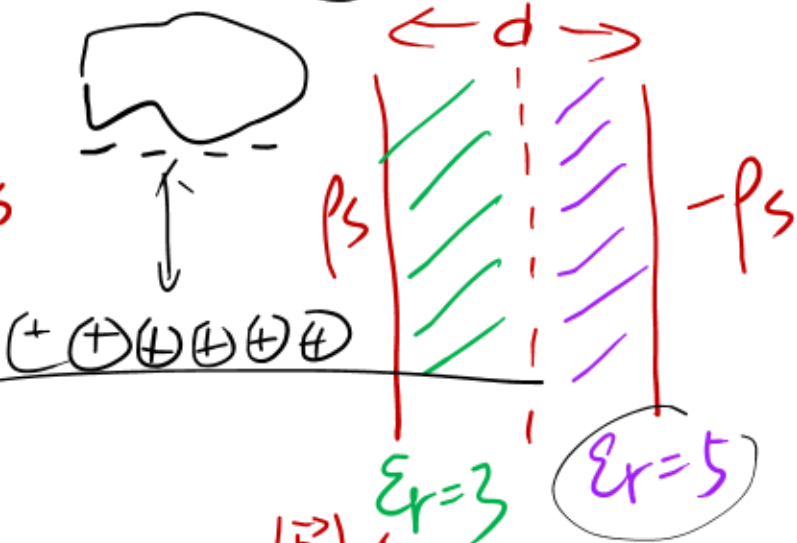
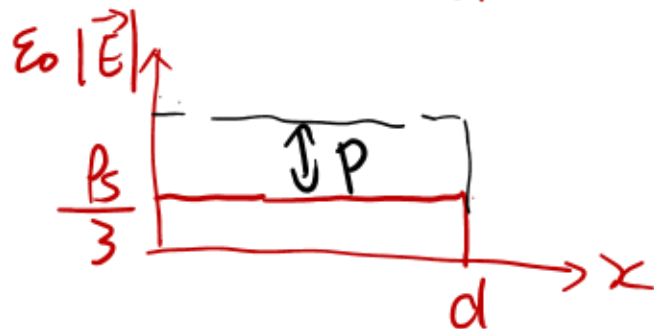
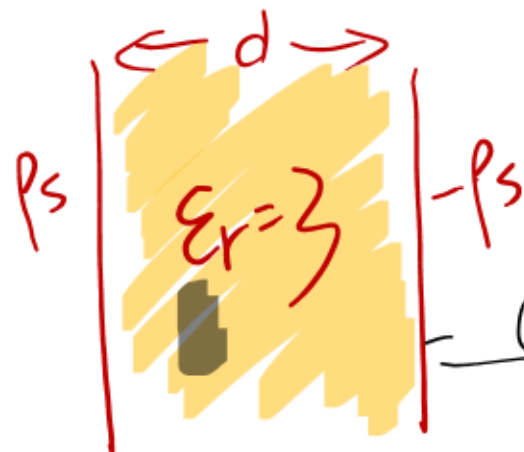
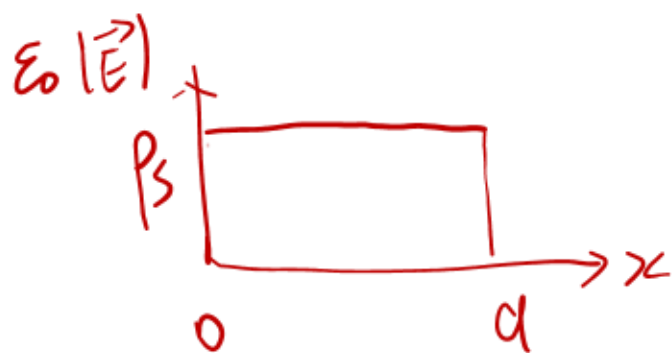
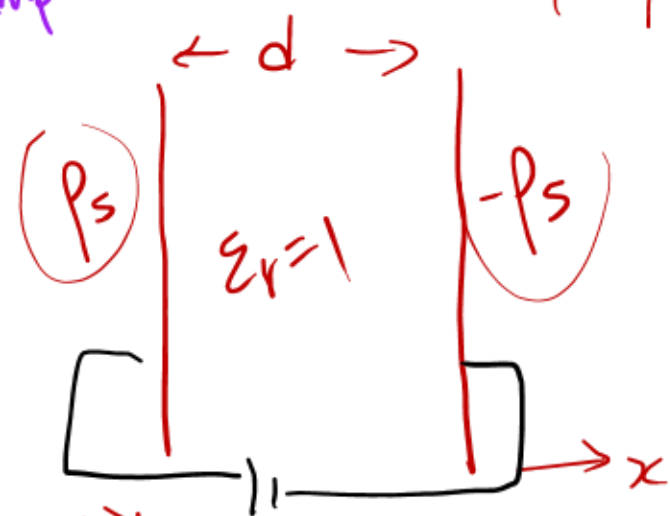
$\epsilon = \epsilon_r \epsilon_0$ and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

Material	Relative Permittivity, ϵ_r	Material	Relative Permittivity, ϵ_r
Vacuum	1	Dry soil	2.5–3.5
Air (at sea level)	1.0006	Plexiglass	3.4
Styrofoam	1.03	Glass	4.5–10
Teflon	2.1	Quartz	3.8–5
Petroleum oil	2.1	Bakelite	5
Wood (dry)	1.5–4	Porcelain	5.7
Paraffin	2.2	Formica	6
Polyethylene	2.25	Mica	5.4–6
Polystyrene	2.6	Ammonia	22
Paper	2–4	Seawater	72–80
Rubber	2.2–4.1	Distilled water	81
^a These are low-frequency values at room temperature (20 °C).			

Example:

$$|\vec{D}| = \rho_s$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$



Dielectric breakdown

Beyond certain dielectric strength E_{ds} , electrons will detach from the molecules and accelerate through the material in the form of a conduction current.

E field strength

When this happens, sparking can occur, and the dielectric material can be permanent damaged. This abrupt change in behavior is called **dielectric breakdown**.

Example: In a parallel-plate capacitor with a separation $d = 0.01 \text{ mm}$ between conducting plates,

If the the capacitor is filled with air, what's the is the breakdown voltage?

$V_{\text{applied}} = 60 \text{ V}$

If the the capacitor is filled with quartz, what's the is the breakdown voltage?

dielectric breakdown

Material	Dielectric Strength, E_{ds} (MV/m)
Air (at sea level)	3
Petroleum oil	12
Polystyrene	20
Glass	25–40
Quartz	30
Bakelite	20
Mica	200

$$V = E_{ds} \cdot d$$

$$\text{air } V = 3 \times 10^6 \text{ V/m} \cdot 0.01 \times 10^{-3} = 30 \text{ V}$$

$$\text{quartz } V = 30 \times 10^6 \text{ V/m} \cdot 0.01 \times 10^{-3} = 300 \text{ V}$$